Non-interacting ants

- 2D Active Brownian particles

- Particle i follows $\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pe}v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \sqrt{2}\mathrm{d}B_t^i \end{cases}$

- $\mathrm{Pe} > 0,\ \sigma_x \geq 0$

- $v(\theta) = (\cos(\theta), \sin(\theta))^\top$ (or, more general, $v \in \mathbb{S}^{d-1}$, ...)

- W_t^i 2D Brownian motion, B_t^i 1D Brownian motion

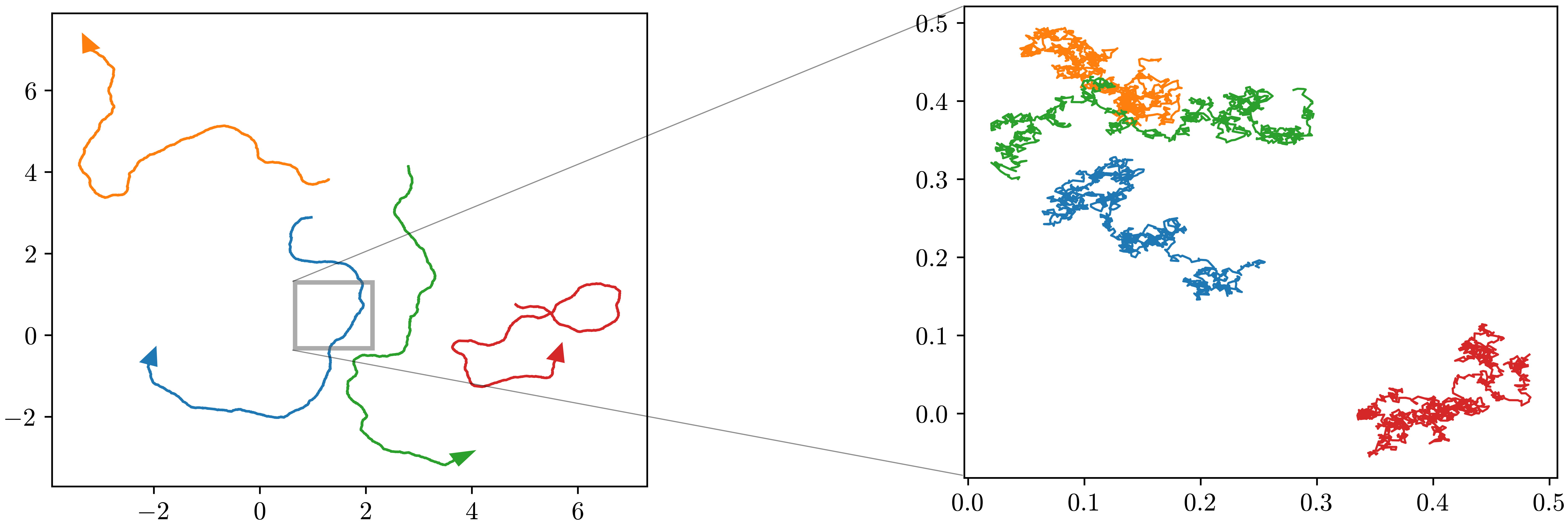
Non-interacting ants

- 2D Active Brownian particles
- Particle i follows $\begin{cases} \mathrm{d}X_t^i & = \mathrm{Pe}\, v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i & = \sqrt{2}\mathrm{d}B_t^i \end{cases}$
- $\mathrm{Pe} > 0,\ \sigma_x \geq 0$
- $v(\theta) = (\cos(\theta), \sin(\theta))^{\mathsf{T}}$ (or, more general, $v \in \mathbb{S}^{d-1}$, …)
- W_t^i 2D Brownian motion, B_t^i 1D Brownian motion

Active Brownian particles

- 2D Active Brownian particles

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pe}\, v(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \sqrt{2}\mathrm{d}B_t^i \end{cases}$$



$\mathrm{Pe} = 2,\ \sigma_x = 10^{-3}$ $\mathrm{Pe} = 0,\ \sigma_x = 10^{-3}$